ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone with radius 3 and height 4 have volumes

A. 36π and 12π

B. 12π and 36π

C. 12π and 4π

D. 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

3201

QUESTION ID

S79-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 36π and 12π

GRADING NOTE ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

True or false?

Two hemispheres of radius r make one sphere of radius r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

3202

QUESTION ID

S79-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

True or false?

A composite volume is the product of the parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

3203

QUESTION ID

S79-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

3205

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

3206

QUESTION ID
S79-F1-Q2

TYPE
Short answer

CORRECT ANSWER
216 − 36π

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

3207

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume scales by $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

A. 1.5

B. 2.25

C. 5

D. 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

3208

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3.375

GRADING NOTE 1.5^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

3209

QUESTION ID

S79-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

GRADING NOTE Any equivalent form. $(4/3\ \pi r^3)/(2\pi r^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

3211

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} km^3

GRADING NOTE Rounding to 3 significant figures required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

3211

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 cm^3.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

3212

QUESTION ID
S79-F2-Q2

TYPE
Short answer

CORRECT ANSWER
7100 cm^3

GRADING NOTE Rounding to the nearest 100 required. r = 75/(2π) ≈ 11.94 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

True or false?

Modeling an orange as a sphere ignores its small bumps.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

3213

QUESTION ID

S79-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Modeling assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

3213

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

A. 28 cm³

B. 113 cm³

C. 339 cm³

D. 36 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

3214

QUESTION ID

S79-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 113 cm³

GRADING NOTE 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

3215

GRADING NOTE Any reasonable answer.

QUESTION ID

S79-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A ball bearing is nearly perfectly round; a pumpkin has ridges, a flattened shape, and a stem, so the model is only approximate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Solve for an unknown dimension from a volume

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Solve a one-step equation with a coefficient	**HSA-REI.B.3**
BACKWARD 2	Square roots and cube roots	**8.EE.A.2**
BACKWARD 3	Rearrange a formula	**HSA-CED.A.4**
FORWARD 1	Find dimensions to meet a target volume	**HSG-GMD.A.3**
FORWARD 2	Design a container that meets constraints	**HSG-MG.A.3**

SET 80 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has volume 360π cm^3 and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

3218

QUESTION ID
S80-M-Q1

TYPE
Short answer

CORRECT ANSWER
10 cm

GRADING NOTE Any equivalent form, with units. $36\pi h = 360\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has volume 96π in^3 and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

3219

QUESTION ID
S80-M-Q2

TYPE
Short answer

CORRECT ANSWER
6 in.

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}\pi r^2 \cdot 8 = 96\pi \rightarrow r^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has volume 36π m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

3220

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 m

GRADING NOTE Any equivalent form, with units. $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

3220

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

3221

QUESTION ID
S80-M-Q4

TYPE
Short answer

CORRECT ANSWER
8 ft

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}s^2 \cdot 12 = 256 \rightarrow s^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

A. 7.1 cm

B. 4.0 cm

C. 2.0 cm

D. 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

3222

QUESTION ID

S80-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 4.0 cm

GRADING NOTE $r = \sqrt{(50/\pi)} \approx 3.99$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $36\pi h = 360\pi$.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

3224

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

3225

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

Solve 4x = 108.

A. 25

B. 104

C. 432

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

3226

QUESTION ID

S80-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE Divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

True or false?

To solve $36\pi h = 360\pi$, dividing both sides by π is allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

3227

QUESTION ID

S80-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\pi \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To solve ⅓x = 12, divide by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

3228

QUESTION ID

S80-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^2 = 36$ for $r > 0$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

3230

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

3231

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt[3]{125} =$

A. 25

B. 5

C. 41.7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

3232

QUESTION ID

S80-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{64} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

3233

QUESTION ID

S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

True or false?

If $r^3 = 8$, then $r = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

3234

QUESTION ID

S80-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $V = \pi r^2 h$ for h.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

3236

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$V/(\pi r^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve V = ⅓Bh for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

3237

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3V/h

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = 4/3\ \pi r^3$ solved for r^3 is

A. $4V/(3\pi)$

B. $V/(4\pi)$

C. $3V/(4\pi)$

D. $3\pi V/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

3238

QUESTION ID

S80-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3V/(4π)

GRADING NOTE Multiply by 3/(4π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $V = \pi r^2 h$ for r needs a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

3239

QUESTION ID

S80-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $r = \sqrt{(V/(\pi h))}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

True or false?

Solving V = Bh for h gives h = VB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

3240

QUESTION ID
S80-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE V/B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A drink can must hold at least 355 cm^3 and have radius 3.3 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

3242

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10.4 cm

GRADING NOTE Form required: a height in tenths that meets the constraint (round UP). $h = 355/(\pi \cdot 3.3^2) = 10.376$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spherical tank must hold at least 1,000 m³. What is the smallest radius, in tenths of a meter, that meets the requirement?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

3243

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.3 m

GRADING NOTE Form required: a radius in tenths that meets the constraint (round UP). $r = \sqrt[3]{(750/\pi)} = 6.2035$; 6.2 m would hold only about 998.3 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

3243

True or false?

To double the volume of a cylinder while keeping its radius, double its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

3244

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Linear in h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

A cone has the same volume as a cylinder of equal radius. The cone's height is

A. 9 times the cylinder's

B. ⅓ of the cylinder's

C. 3 times the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

3245

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3 times the cylinder's

GRADING NOTE $\frac{1}{3}\pi r^2 h_1 = \pi r^2 h_2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

3246

QUESTION ID

S80-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Rounding to 0.1 required. $\sqrt[3]{(36\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

3246

SET 80 • HSG-GMD.A.3

FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical vase must hold 2,000 cm³ and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

3248

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: r = √(2000/(20π)) ≈ 5.6 cm, which is at least 5 cm.

GRADING NOTE Form required: the conclusion with the computed radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A conical paper cup must hold at least 150 cm^3 and have radius 3.5 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

3249

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

11.7 cm

GRADING NOTE Form required: a height in tenths that meets the constraint (round UP). $h = 450/(\pi \cdot 3.5^2) = 11.693$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

True or false?

A container's volume requirement fixes each of its dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

3250

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It fixes only a relationship: many (r, h) pairs give the same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold 100π cm³. Which (r, h) works?

A. (5, 4)

B. (5, 20)

C. (4, 5)

D. (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

3251

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (5, 4)

GRADING NOTE $25 \cdot 4 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give two different cylinders (radius and height) that each hold 64π cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

3252

QUESTION ID
S80-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example r = 4, h = 4 and r = 2, h = 16.

GRADING NOTE Any two (r, h) pairs with $r^2h = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify cross-sections of three-dimensional objects

MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Cross-sections of right rectangular prisms and pyramids	**7.G.A.3**
BACKWARD 2	Properties of polygons	**5.G.B.4**
BACKWARD 3	Parts of solids	**6.G.A.4**
FORWARD 1	Areas of cross-sections	**HSG-GMD.B.4**
FORWARD 2	Use cross-section areas to compare volumes (Cavalieri)	**HSG-GMD.A.2**

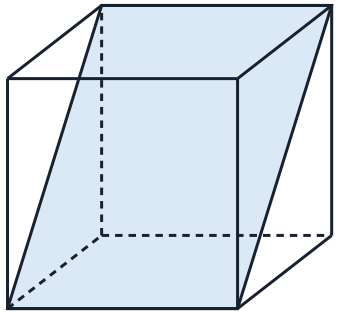
SET 81 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

3255

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle (not a square)

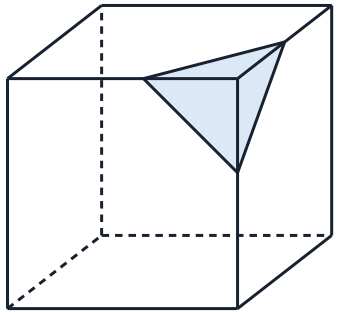
GRADING NOTE Any equivalent answer. Two sides are cube edges (s) and two are face diagonals (s√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

3256

QUESTION ID

S81-M-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle

GRADING NOTE Any equivalent answer. (Equilateral if the three cut points are equally far from the corner.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A right circular cone is cut by a vertical plane through its apex. The cross-section is

A. a circle

B. a rectangle

C. a triangle (isosceles)

D. an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

3257

QUESTION ID

S81-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a triangle (isosceles)

GRADING NOTE Two slant edges and a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

3258

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

(a) circle (b) rectangle (c) ellipse

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius 10 is cut by a plane 6 units from its center. Name the shape of the cross-section and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

3259

QUESTION ID

S81-M-Q5

TYPE

Short answer

CORRECT ANSWER

A circle of radius 8

GRADING NOTE Form required: the shape and the radius. $\sqrt{(10^2 - 6^2)} = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

3261

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

3262

QUESTION ID

S81-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A smaller square

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

A. trapezoid

B. square

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

3263

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. triangle

GRADING NOTE Through the apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

3264

QUESTION ID

S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent to the face.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

A cut of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

3265

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3267

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

True or false?

Every square is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

3268

QUESTION ID

S81-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hierarchy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

A polygon with six sides is a

A. hexagon

B. pentagon

C. octagon

D. heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

3269

QUESTION ID

S81-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. hexagon

GRADING NOTE Name.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

3270

QUESTION ID

S81-B2-Q4

TYPE

Short answer

CORRECT ANSWER

isosceles

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

True or false?

A trapezoid has no parallel sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

3271

QUESTION ID

S81-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At least one pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

3273

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

3274

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

How many edges meet at each vertex of a cube?

A. 6

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

3275

QUESTION ID

S81-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Corner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

True or false?

A cylinder has two congruent circular bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

3276

QUESTION ID

S81-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

True or false?

A sphere has one edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

3277

QUESTION ID

S81-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

3279

QUESTION ID
S81-F1-Q1

TYPE
Short answer

CORRECT ANSWER
16√2

GRADING NOTE Form required: exact value. 4 × 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

3280

QUESTION ID

S81-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius at that level = 6 · (12 − 4)/12 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q2**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

3281

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3-4-5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

A. octagon

B. pentagon

C. heptagon

D. hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

3282

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. hexagon

GRADING NOTE Plane through the midpoints of six edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 10 and height 15. Find the exact area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

3283

QUESTION ID
S81-F1-Q5

TYPE
Short answer

CORRECT ANSWER
100/9 (about 11.1)

GRADING NOTE Form required: exact value or rounding to 0.1. Side 10 · 5/15 = 10/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

3285

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 70 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

3286

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

True or false?

Cavalieri's principle can compare two solids of different heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

3287

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solids must have equal heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

A hemisphere of radius r rests on its flat base. At height y above the base plane ($0 \le y \le r$), its horizontal cross-section has area

A. $\pi(r^2 - y^2)$

B. $\pi(r - y)^2$

C. πr^2

D. πy^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

3288

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $\pi(r^2 - y^2)$

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

3289

GRADING NOTE Any reasonable answer.

QUESTION ID

S81-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Volume can be built from thin slices; equal slice areas at each level give equal volumes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify three-dimensional objects generated by rotations of two-dimensional objects

MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Rotation of a point about a center	**HSG-CO.A.4**
BACKWARD 2	Cross-sections perpendicular to the axis	**HSG-GMD.B.4**
BACKWARD 3	Names of solids	**6.G.A.4**
FORWARD 1	Volumes of solids of rotation	**HSG-GMD.B.4**
FORWARD 2	Design an object as a solid of rotation	**HSG-MG.A.3**

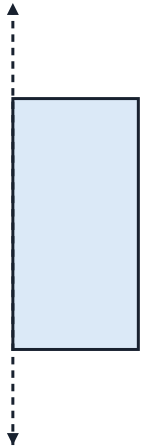
SET 82 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

3292

QUESTION ID

S82-M-Q1

TYPE

Short answer

CORRECT ANSWER

A cylinder

GRADING NOTE Any equivalent answer.

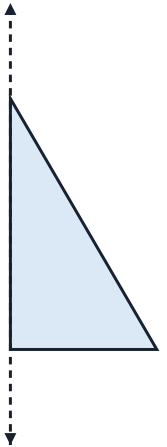
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The right triangle is rotated 360° about the dashed line. Name the solid formed.

Solid: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3293

QUESTION ID

S82-M-Q2

TYPE

Short answer

CORRECT ANSWER

A cone

GRADING NOTE Any equivalent answer.

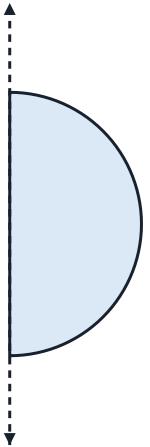
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3293

ANSWER KEY Reduced copy of the question page, with its answer

The semicircle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

3294

QUESTION ID

S82-M-Q3

TYPE

Short answer

CORRECT ANSWER

A sphere

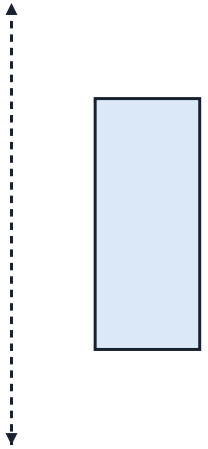
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3295

QUESTION ID

S82-M-Q4

TYPE

Short answer

CORRECT ANSWER

A hollow cylinder (a cylinder with a cylindrical hole through the middle, like a pipe)

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

A right triangle is rotated about its hypotenuse. The solid formed is

A. a sphere

B. one cone

C. two cones joined at their bases

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

3296

QUESTION ID

S82-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. two cones joined at their bases

GRADING NOTE Each leg sweeps a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

True or false?

In a plane, rotating a point P (P different from the center O) 360° about O traces a circle with center O and radius OP.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

3298

QUESTION ID

S82-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every image is OP from O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

3299

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A circle of radius 3 cm

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

A point ON the axis of rotation

A. traces a line

B. stays fixed

C. traces a circle

D. disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3300

QUESTION ID

S82-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. stays fixed

GRADING NOTE Distance 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3300

ANSWER KEY Reduced copy of the question page, with its answer

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

3301

QUESTION ID

S82-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The curved side of a cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

True or false?

Points farther from the axis trace smaller circles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

3302

QUESTION ID

S82-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Larger circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

3304

QUESTION ID
S82-B2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every point traces a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

3305

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

A pipe cut perpendicular to its axis gives a

A. rectangle

B. ring (annulus)

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

3306

QUESTION ID

S82-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ring (annulus)

GRADING NOTE Hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

3307

QUESTION ID

S82-B2-Q4

TYPE

Short answer

CORRECT ANSWER

A circle (a great circle)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

True or false?

A cylinder cut perpendicular to its axis gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

3308

QUESTION ID
S82-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

3310

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

3310

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

3311

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Cone

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

A. sphere

B. cone

C. circle

D. cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

3312

QUESTION ID

S82-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. sphere

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

True or false?

A cone has one flat face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

3313

QUESTION ID

S82-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

True or false?

A sphere has a flat base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

3314

QUESTION ID

S82-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

3316

QUESTION ID
S82-F1-Q1

TYPE
Short answer

CORRECT ANSWER
45π

GRADING NOTE Form required: exact value. r = 3, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

3316

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

3317

QUESTION ID
S82-F1-Q2

TYPE
Short answer

CORRECT ANSWER
75π

GRADING NOTE Form required: exact value. r = 5, h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

Rotating one rectangle about two different sides can give solids with different volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

3318

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A 3-by-5 rectangle gives 45π about its 5-unit side and 75π about its 3-unit side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

A. 48π

B. 9π

C. 12π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

3319

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 16π

GRADING NOTE r = 4, h = 3: ⅓π · 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

3320

QUESTION ID
S82-F1-Q5

TYPE
Short answer

CORRECT ANSWER
36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

3322

QUESTION ID
S82-F2-Q1

TYPE
Short answer

CORRECT ANSWER
40π cm^3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

3323

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

3323

True or false?

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

3324

QUESTION ID

S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That gives a cone; rotate a quarter-circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

To make a ring-shaped washer by rotation, rotate a rectangle about a line in the rectangle's plane, parallel to one of its sides, that

A. passes through the rectangle's center

B. contains one side

C. lies outside the rectangle

D. cuts through the rectangle off-center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

3325

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. lies outside the rectangle

GRADING NOTE The gap between the axis and the rectangle becomes the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line in the rectangle's plane, parallel to its long side and 2 cm from it, outside the rectangle. Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

3326

QUESTION ID
S82-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1,200π cm³

GRADING NOTE Form required: exact value. Outer radius 8, inner radius 2: π(64 − 4)(20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

Model objects with geometric shapes

MAIN	Geometry main question	**HSG-MG.A.1**
BACKWARD 1	Volume formulas	**HSG-GMD.A.3**
BACKWARD 2	Radius from a circumference	**7.G.B.4**
BACKWARD 3	Convert cubic units	**6.RP.A.3.d**
FORWARD 1	Judge a model against a measurement	**HSG-MG.A.1**
FORWARD 2	Use the model to estimate mass with density	**HSG-MG.A.2**

SET 83 • HSG-MG.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

3329

QUESTION ID

S83-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

About 2.3 m³

GRADING NOTE REQUIRED WORK: Find the radius from the circumference, then use $V = \pi r^2 h$.

Rounding to 0.1 required. $r = 2.2/(2\pi) \approx 0.35$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

3330

QUESTION ID

S83-M-Q2

TYPE

Short answer

CORRECT ANSWER

50.3 m^3

GRADING NOTE Rounding to 0.1 required. 16π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

Which shape is the best model for a soup can?

A. Cone

B. Cylinder

C. Square pyramid

D. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

3331

QUESTION ID

S83-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Cylinder

GRADING NOTE Two circular bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 m^3 = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

3332

QUESTION ID

S83-M-Q4

TYPE

Short answer

CORRECT ANSWER

450,000 L

GRADING NOTE Any equivalent form. 450 m^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

3333

QUESTION ID
S83-M-Q5

TYPE
Short answer

CORRECT ANSWER
For example: height and the width at the bottom of the branches; the real tree has gaps, a trunk, and an uneven outline.

GRADING NOTE Any reasonable measurement and limitation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

3335

QUESTION ID

S83-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

3336

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 4 prism?

A. 52

B. 9

C. 26

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

3337

QUESTION ID

S83-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 24

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

V = ⅓πr²h is the cone formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

3338

QUESTION ID

S83-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sphere's volume is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

3339

QUESTION ID

S83-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4/3\ \pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

3341

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

3342

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.35 m

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

r =

A. C/(2π)

B. πC

C. 2πC

D. C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

3343

QUESTION ID

S83-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C/(2π)

GRADING NOTE C = 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

True or false?

A tape measure around a trunk gives its circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

3344

QUESTION ID
S83-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distance around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The radius is C/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

3345

QUESTION ID

S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many liters are in 2 m^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

3347

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 L

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

3348

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1,000 cm^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

450 m^3 = ___ L

A. 45,000

B. 0.45

C. 450,000

D. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

3349

QUESTION ID

S83-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 450,000

GRADING NOTE × 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

True or false?

$1\ \text{mL} = 1\ \text{cm}^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

3350

QUESTION ID

S83-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1\ m^3 = 100\ L$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

3351

QUESTION ID

S83-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1,000 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

3353

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 771 mL; about 2.8% too high

GRADING NOTE Form required: both (rounding to 1 mL and 0.1%). Percent error = (model − actual)/actual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

3354

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

126π m³

GRADING NOTE Form required: exact value. 108π + 18π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

True or false?

A model that overestimates by 3% may still be useful for planning.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

3355

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

A football (pointed ends) is best modeled by

A. two cones joined at their bases

B. a cube

C. a sphere

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

3356

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. two cones joined at their bases

GRADING NOTE Pointed ends; other answers with reasons may be discussed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

3357

QUESTION ID

S83-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The real bottle narrows at the neck and has thick walls, so it holds less than the full cylinder.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk modeled as a cylinder has volume about 2.3 m³. The wood has density 600 kg/m³. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

3359

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

About 1,380 kg

GRADING NOTE Rounding to the nearest 10 kg required. 2.3 × 600 = 1,380 (using the unrounded volume 2.311 m³ gives 1390 kg, also accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sand pile modeled as a cone has volume 16π m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3360

QUESTION ID
S83-F2-Q2

TYPE
Short answer

CORRECT ANSWER
80 t

GRADING NOTE Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3360

True or false?

A more accurate volume model gives a more accurate mass estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

3361

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

A. 450,000 kg

B. 450 kg

C. 45,000 kg

D. 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

3362

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000 kg

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

3363

QUESTION ID

S83-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The model gives the volume; mass = density × volume.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on area

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Unit rates	**6.RP.A.3.b**
BACKWARD 2	Areas of rectangles and circles	**7.G.B.6**
BACKWARD 3	Compare ratios	**7.RP.A.2**
FORWARD 1	Convert area units in density problems	**HSG-MG.A.2**
FORWARD 2	Plan capacity from an area density	**HSG-MG.A.3**

SET 84 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

3366

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

7,000 people per square mile

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

3367

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

159 trees per km²

GRADING NOTE Rounding to the nearest whole number required. Area $9\pi \approx 28.27$ km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

3368

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

Town B (450 per km² vs 400 per km²)

GRADING NOTE Form required: the conclusion with both densities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

3369

QUESTION ID
S84-M-Q4

TYPE
Short answer

CORRECT ANSWER
6,000 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a unit of population density based on area?

A. people per mile

B. people per square mile

C. square miles per person per year

D. people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

3370

QUESTION ID

S84-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. people per square mile

GRADING NOTE Count ÷ area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

3372

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

3373

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$5 per m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Unit rate for 300 people on 4 km²?

A. 0.013 km² per person

B. 1,200 people per km²

C. 296 people per km²

D. 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

3374

QUESTION ID

S84-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 75 people per km²

GRADING NOTE 300 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

True or false?

A unit rate compares a quantity to one unit of another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

3375

QUESTION ID

S84-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

True or false?

600 people on 3 km² is 1,800 people per km².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

3376

QUESTION ID

S84-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

3378

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

24 km²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

3379

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9π km²

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square region 5 mi on a side has area

A. 20 mi²

B. 10 mi²

C. 5 mi²

D. 25 mi²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3380

QUESTION ID

S84-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25 mi²

GRADING NOTE 5^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3380

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Area units are square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

3381

QUESTION ID

S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two dimensions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

True or false?

A circle with radius 3 has area 6π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

3382

QUESTION ID

S84-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

3384

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9,000/20 (450 > 400)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

True or false?

To compare densities, compare the unit rates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

3385

QUESTION ID

S84-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rate is greatest?

A. 80 per 4 km²

B. 50 per 2 km²

C. 60 per 3 km²

D. 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

3386

QUESTION ID

S84-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50 per 2 km²

GRADING NOTE 25, 20, 20, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

3386

ANSWER KEY Reduced copy of the question page, with its answer

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

3387

QUESTION ID

S84-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A (25 per acre vs 20)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The town with more people is always denser.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

3388

QUESTION ID

S84-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Depends on area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3390

QUESTION ID
S84-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2703 people per km²

GRADING NOTE Rounding to the nearest whole number required. Divide by 2.59.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3390

ANSWER KEY Reduced copy of the question page, with its answer

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

3391

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14 kg

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A density in people per km² is a larger number than the same density in people per mi².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

3392

QUESTION ID
S84-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A km² is smaller, so fewer people per km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

A farm plants 2 trees per 10 m². How many trees on 1 hectare (10,000 m²)?

A. 2,000

B. 200

C. 5,000

D. 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3393

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2,000

GRADING NOTE 0.2 per m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3393

ANSWER KEY Reduced copy of the question page, with its answer

A park of 3 hectares (1 ha = 10,000 m²) has 5,400 visitors. Visitors per 100 m²?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3394

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. 300 units of 100 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3394

SET 84 • HSG-MG.A.2

FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Safety rules allow 2 people per m^2 in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

3396

QUESTION ID
S84-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2,400 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest whole number of square meters that meets the limit?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

3397

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3,334 m²

GRADING NOTE Form required: a whole number of m² that meets the limit. 5,000 ÷ 1.5 = 3,333.3…, so round UP to 3,334 (3,333 m² would allow only 4,999.5 people).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

True or false?

When planning for a safety limit, round the capacity up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

3398

QUESTION ID
S84-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Round capacity down (and area up) to stay within the limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

A. 235 people

B. 471 people

C. 75 people

D. 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

3399

QUESTION ID

S84-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 235 people

GRADING NOTE 75π ≈ 235.6, round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular dance floor of radius 5 m allows at most 3 people per m², and $3 \cdot 25\pi \approx 235.6$. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

3400

QUESTION ID

S84-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Rounding up would exceed the allowed density.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on volume

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Volume of prisms and cylinders	**HSG-GMD.A.3**
BACKWARD 2	Unit rates	**6.RP.A.3.b**
BACKWARD 3	Products of numbers in scientific notation	**8.EE.A.4**
FORWARD 1	Use density to identify materials and find volumes	**HSG-MG.A.2**
FORWARD 2	Design within a mass limit	**HSG-MG.A.3**

SET 85 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

3403

QUESTION ID
S85-M-Q1

TYPE
Short answer

CORRECT ANSWER
2.7 g/cm³

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

3404

QUESTION ID

S85-M-Q2

TYPE

Short answer

CORRECT ANSWER

1571 kg

GRADING NOTE Rounding to the nearest kilogram required. 0.5π m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

3405

QUESTION ID

S85-M-Q3

TYPE

Short answer

CORRECT ANSWER

28,800 BTU

GRADING NOTE Any equivalent form. 1,440 ft^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sample has 3×10^6 bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

3406

QUESTION ID

S85-M-Q4

TYPE

Short answer

CORRECT ANSWER

7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Water has density 1 g/cm³. Which object floats in water?

A. A 1,000 g brick with volume 400 cm³

B. A 600 g block with volume 500 cm³

C. A 100 g ball with volume 50 cm³

D. A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

3407

QUESTION ID

S85-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. A 400 g block with volume 500 cm³

GRADING NOTE Density less than 1 g/cm³ (0.8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

3409

QUESTION ID

S85-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3410

QUESTION ID

S85-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.5π m³

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3410

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 12 × 15 × 8 ft room?

A. 1,440 ft³

B. 180 ft³

C. 2,880 ft³

D. 35 ft³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

3411

QUESTION ID

S85-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1,440 ft³

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

True or false?

Volume is needed before a volume density can be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

3412

QUESTION ID
S85-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Density = amount ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 m cube has volume 3 m³.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

3413

QUESTION ID

S85-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

81 g in 30 cm³: grams per cm³?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

3415

QUESTION ID

S85-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2.7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

20 BTU per ft^3 for 100 ft^3: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

3416

QUESTION ID

S85-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2,000 BTU

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

500 g in 250 cm³ is

A. 250 g/cm³

B. 2 g/cm³

C. 0.5 g/cm³

D. 750 g/cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

3417

QUESTION ID

S85-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 g/cm³

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

True or false?

Amount = rate × quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

3418

QUESTION ID

S85-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rate use.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

True or false?

Density = volume ÷ mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

3419

QUESTION ID

S85-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

3421

QUESTION ID

S85-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

3422

QUESTION ID

S85-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.4×10^6

GRADING NOTE Form required: scientific notation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2 \times 10^3)(4 \times 10^2) =$

A. 6×10^5

B. 8×10^5

C. 8×10^6

D. 8×10^1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

3423

QUESTION ID

S85-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8×10^5

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$10^3 \times 10^2 = 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

3424

QUESTION ID

S85-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Exponent rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5 \times 10^2 \times 3 = 15 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

3425

QUESTION ID

S85-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.5×10^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

3427

QUESTION ID

S85-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes:

9.65 ÷ 0.5 = 19.3 g/cm³

GRADING NOTE Form required: the conclusion with the density.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

3428

QUESTION ID

S85-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.9 cm

GRADING NOTE Rounding to 0.1 required. Volume 500 cm³; edge $\sqrt[3]{500}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

True or false?

Two objects with equal mass but different volumes have different densities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

3429

QUESTION ID

S85-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density = m/V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

A. 1,800 g

B. 2,000 g

C. 2,222 g

D. 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

3430

QUESTION ID

S85-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1,800 g

GRADING NOTE 0.9 × 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A balloon holds 0.02 m^3 of helium (0.18 kg/m^3). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

3431

QUESTION ID
S85-F1-Q5

TYPE
Short answer

CORRECT ANSWER
3.6 g

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

3433

QUESTION ID

S85-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

3434

QUESTION ID
S85-F2-Q2

TYPE
Short answer

CORRECT ANSWER
50,000 cm³

GRADING NOTE Any equivalent form. 40,000 g ÷ 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

3434

True or false?

Mass limits become volume limits by multiplying by the density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

3435

QUESTION ID

S85-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by the density: V = m/d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

A. 10 m

B. 3.14 m

C. 1 m

D. 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

3436

QUESTION ID
S85-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 1 m

GRADING NOTE 3.14 m³ ÷ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

3436

ANSWER KEY Reduced copy of the question page, with its answer

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

3437

QUESTION ID

S85-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 0.012 m³ × 2,400 kg/m³ = 28.8 kg, which exceeds 25 kg.

GRADING NOTE Form required: the conclusion with the mass.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to satisfy physical constraints

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Volume of a rectangular prism	**7.G.B.6**
BACKWARD 2	Use the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Ratios and rates	**7.RP.A.2**
BACKWARD 4	Inequalities as limits	**7.EE.B.4**
FORWARD 1	Satisfy several constraints at once	**HSG-MG.A.3**
FORWARD 2	Find the best design with a function graph	**HSF-IF.C.7.c**

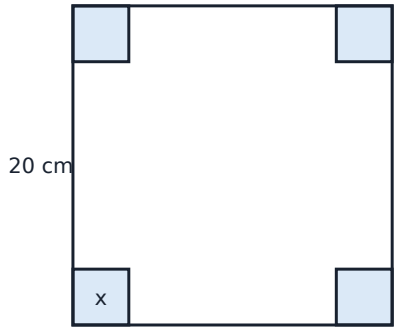
SET 86 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?



Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

3440

QUESTION ID

S86-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2: 512 cm³; x = 3: 588 cm³; x = 4: 576 cm³; x = 3 is largest

GRADING NOTE Form required: the three volumes and the choice. $V = x(20 - 2x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

3441

QUESTION ID
S86-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 ft

GRADING NOTE Any equivalent form. 30 × 12 = 360 in.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

3442

QUESTION ID

S86-M-Q3

TYPE

Short answer

CORRECT ANSWER

The 10 m by 10 m pen (100 m² vs 75 m²)

GRADING NOTE Form required: the choice with both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

3443

QUESTION ID

S86-M-Q4

TYPE

Short answer

CORRECT ANSWER

9.4 m³

GRADING NOTE Rounding to 0.1 required. Radius 1 m, height 3 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

3443

ANSWER KEY Reduced copy of the question page, with its answer

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

A. 18 ft

B. 12 ft

C. 13 ft

D. 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

3444

QUESTION ID

S86-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 12 ft

GRADING NOTE $\sqrt{(169 - 25)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

3446

QUESTION ID

S86-B1-Q1

TYPE

Short answer

CORRECT ANSWER

588 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

3447

QUESTION ID

S86-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a box 12 by 12 by 4?

A. 28

B. 576

C. 48

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

3448

QUESTION ID

S86-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 576

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

True or false?

Cutting larger corners makes the box taller but its base smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

3449

QUESTION ID

S86-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Trade-off.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

True or false?

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is 20 − x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

3450

QUESTION ID
S86-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 20 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

3452

QUESTION ID

S86-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

3453

QUESTION ID

S86-B2-Q2

TYPE

Short answer

CORRECT ANSWER

361 in.

GRADING NOTE Rounding to the nearest inch required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

A. right triangle

B. circle

C. square

D. equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

3454

QUESTION ID

S86-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. right triangle

GRADING NOTE Wall ⊥ ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

True or false?

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

3455

QUESTION ID

S86-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite the right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

True or false?

The ladder's height on the wall is 13 + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

3456

QUESTION ID

S86-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

3458

QUESTION ID

S86-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

3459

QUESTION ID

S86-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 3 : 30 = 1 : 10, and 1/10 > 1/12.

GRADING NOTE Form required: yes/no with the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

360 in. =

A. 30 ft

B. 36 ft

C. 12 ft

D. 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

3460

GRADING NOTE ÷ 12.

QUESTION ID

S86-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

3461

QUESTION ID

S86-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

3462

QUESTION ID

S86-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

3464

QUESTION ID

S86-B4-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

3465

QUESTION ID

S86-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

"At most 3 m" means

A. ≥ 3

B. < 3 only

C. ≤ 3

D. = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

3466

QUESTION ID

S86-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ≤ 3

GRADING NOTE Inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At least 5 ft" allows exactly 5 ft.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

3467

QUESTION ID

S86-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ≥.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At most 12" excludes 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

3468

QUESTION ID

S86-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Includes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm³ AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

3470

QUESTION ID

S86-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 and x = 3

GRADING NOTE Form required: both values. Volumes 512 and 588; x = 4, 5 are too tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

3471

QUESTION ID

S86-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it needs 24 ft of run.

GRADING NOTE Form required: the conclusion with the run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

True or false?

A design that satisfies one constraint may fail another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

3472

QUESTION ID

S86-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Check all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

A. 6 m wide by 8 m long

B. 4 m wide by 10 m long

C. 7 m wide by 7 m long

D. 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

3473

QUESTION ID

S86-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6 m wide by 8 m long

GRADING NOTE Width 6, area 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder 3 m tall must hold at least 10 m³ and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

3474

QUESTION ID

S86-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the largest is $r = 1$ m, giving $3\pi \approx 9.4$ m³ < 10 m³.

GRADING NOTE Form required: the conclusion with the volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

3474

SET 86 • HSG-MG.A.3

FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

3476

QUESTION ID

S86-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$V(x) = x(20 - 2x)^2$, $0 < x < 10$

GRADING NOTE Form required: the function and the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

3477

QUESTION ID

S86-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \approx 3.3$; $V \approx 593\ cm^3$

GRADING NOTE Form required: both. (Exact: x = 10/3, V = 16,000/27 ≈ 592.6.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

True or false?

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

3478

QUESTION ID
S86-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The zeros are $x = 0$ and $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

A. the cut that makes no box

B. the base area

C. the largest volume

D. the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

3479

QUESTION ID
S86-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. the largest volume

GRADING NOTE Maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

3480

QUESTION ID
S86-F2-Q5

TYPE
Short answer

CORRECT ANSWER
V(5) = 500 < 593; the maximum of a cubic need not be at the middle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to minimize cost

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Surface area of prisms and cylinders	**7.G.B.6**
BACKWARD 4	Areas of surfaces with openings	**7.G.B.6**
BACKWARD 3	Unit prices and rounding up purchases	**7.EE.B.3**
FORWARD 1	Compare designs with equal volume for least material	**HSG-MG.A.3**
FORWARD 2	Decide between designs using expected cost	**HSS-MD.B.7**

SET 87 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can with radius 4 cm and height 10 cm is made of metal costing \$0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

3483

QUESTION ID
S87-M-Q1

TYPE
Short answer

CORRECT ANSWER
112π cm² ≈ 351.9 cm²; cost \$0.70

GRADING NOTE Form required: the area and the cost. $2\pi r^2 + 2\pi rh = 32\pi + 80\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

3484

QUESTION ID
S87-M-Q2

TYPE
Short answer

CORRECT ANSWER
Both hold 160π cm³; A uses 112π cm², B uses 114π cm², so A uses less metal.

GRADING NOTE Form required: both volumes, both areas, and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

3485

QUESTION ID
S87-M-Q3

TYPE
Short answer

CORRECT ANSWER
$32

GRADING NOTE Any equivalent form. 34.2 m² needs 3.42 L, so buy 4 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two closed cardboard boxes each hold 600 cm³: box A is 10 cm by 10 cm by 6 cm and box B is 5 cm by 10 cm by 12 cm. Cardboard costs $0.01 per cm² (ignore thickness and overlaps). Find the cost of each box and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

3486

QUESTION ID

S87-M-Q4

TYPE

Short answer

CORRECT ANSWER

A: 440 cm², $4.40; B: 460 cm², $4.60; box A is cheaper.

GRADING NOTE Form required: both surface areas, both costs, and the choice. A: 2(100 + 60 + 60); B: 2(50 + 60 + 120).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 4 m by 3.5 m floor is covered with 0.5 m square tiles that cost $3.25 each and are sold individually. Plan A uses these tiles; plan B uses sheet flooring at $11 per m². Find the cost of each plan and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

3487

GRADING NOTE Form required: both costs and the choice.

QUESTION ID

S87-M-Q5

TYPE

Short answer

CORRECT ANSWER

A: 8 × 7 = 56 tiles, $182; B: 14 m² × $11 = $154; plan B is cheaper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

3489

QUESTION ID

S87-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3490

QUESTION ID

S87-B1-Q2

TYPE

Short answer

CORRECT ANSWER

80π

GRADING NOTE Form required: exact value. 2πrh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3490

Surface area of a 2 × 3 × 4 box?

A. 24

B. 48

C. 52

D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

3491

QUESTION ID

S87-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 52

GRADING NOTE 2(6 + 8 + 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

True or false?

The side of a cylinder unrolls into a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

3492

QUESTION ID

S87-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Net.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

True or false?

A cylinder's surface area is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

3493

QUESTION ID

S87-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 4

Areas of surfaces with openings • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a 1.5 m by 1.2 m window. Area to paint?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q1/5

3495

QUESTION ID

S87-B4-Q1

TYPE

Short answer

CORRECT ANSWER

34.2 m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 4 m by 3.5 m floor: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q2/5

3496

QUESTION ID

S87-B4-Q2

TYPE

Short answer

CORRECT ANSWER

14 m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q2/5

A 5 m by 2.4 m wall has a 0.9 m by 2 m door. Paintable area?

A. 10.2 m²

B. 12 m²

C. 1.8 m²

D. 13.8 m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q3/5

3497

QUESTION ID

S87-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10.2 m²

GRADING NOTE 12 − 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q3/5

True or false?

Openings in a surface are subtracted from its area before buying material.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q4/5

3498

QUESTION ID

S87-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No material is needed there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q4/5

True or false?

A floor of 14 m² needs 14 tiles that are 0.5 m on a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q5/5

3499

QUESTION ID

S87-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each tile is 0.25 m², so 56 tiles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3501

QUESTION ID

S87-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

3502

QUESTION ID

S87-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$0.70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

A. $30

B. $25.20

C. $24

D. $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3503

QUESTION ID

S87-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $30

GRADING NOTE Buy 5 yd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When items are sold in whole units, round the amount needed up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

3504

QUESTION ID

S87-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Enough material.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3.1 cans of paint can be bought as 3 cans.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3505

QUESTION ID

S87-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Need 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3505

SET 87 • HSG-MG.A.3

FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A closed box with a square base must hold 1,000 cm^3 (ignore the thickness of the material). Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

3507

QUESTION ID
S87-F1-Q1

TYPE
Short answer

CORRECT ANSWER
8 cm: 628 cm^2; 10 cm: 600 cm^2; 12 cm: about 621 cm^2; the 10 cm cube uses the least

GRADING NOTE Form required: the three areas and the choice. $h = 1000/s^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A closed cylinder must hold 1,000 cm³. Compare its surface area (to the nearest cm²) for radii 4 cm, 5.42 cm and 7 cm, and choose the radius that uses the least material.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

3508

QUESTION ID

S87-F1-Q2

TYPE

Short answer

CORRECT ANSWER

r = 4: 601 cm²;
r = 5.42: 554 cm²;
r = 7: 594 cm²;
r = 5.42 cm uses the least.

GRADING NOTE Form required: three areas and the choice. h = 1000/(πr²), so the area is 2πr² + 2000/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

True or false?

Among closed boxes with a square base and a fixed volume (ignoring material thickness), the cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

3509

QUESTION ID
S87-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Classic result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

For a closed cylinder with a fixed volume, the least surface area occurs when

A. the height equals the radius

B. the height is as large as possible

C. the radius is as large as possible

D. the height equals the diameter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

3510

QUESTION ID

S87-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. the height equals the diameter

GRADING NOTE h = 2r (r = 5.42 cm gives h ≈ 10.84 cm).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

3510

ANSWER KEY Reduced copy of the question page, with its answer

Give one reason a manufacturer might NOT choose the least-material can shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

3511

QUESTION ID
S87-F1-Q5

TYPE
Short answer

CORRECT ANSWER
For example: it is hard to grip, does not fit standard shelves, or leaves too little label space.

GRADING NOTE Any reasonable reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

3513

QUESTION ID
S87-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A: $1.20 + 0.02 × $20 = $1.60; B: $0.90 + 0.08 × $20 = $2.50; choose A

GRADING NOTE Form required: both expected costs and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The standard metal for a can never fails. A cheaper metal fails 5% of the time, and each failure costs $40 to replace. By how much must the cheaper metal's price per can be below the standard price for its expected cost per can to be strictly lower?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

3514

QUESTION ID
S87-F2-Q2

TYPE
Short answer

CORRECT ANSWER
By more than $2.00 (0.05 × $40 = $2.00 expected failure cost)

GRADING NOTE Form required: "more than $2.00"; exactly $2.00 cheaper gives equal expected costs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

True or false?

The cheapest material always gives the cheapest design.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

3515

QUESTION ID
S87-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Failures and replacement costs can make it more expensive overall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

A. upfront cost × probability

B. upfront cost + probability of failure × cost of failure

C. cost of failure only

D. upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

3516

QUESTION ID
S87-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. upfront cost + probability of failure × cost of failure

GRADING NOTE Weighted cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

3517

QUESTION ID
S87-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example shipping weight, labor, breakage, or recycling.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 105 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design typographic grid systems

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Proportional reasoning with ratios	**7.RP.A.2**
BACKWARD 2	Operate with decimals	**6.NS.B.3**
BACKWARD 3	Factors of a whole number	**4.OA.B.4**
FORWARD 1	Choose grids that support several layouts	**HSG-MG.A.3**
FORWARD 2	Scale a grid layout to a new page size	**HSG-SRT.A.2**

SET 105 • HSG-MG.A.3

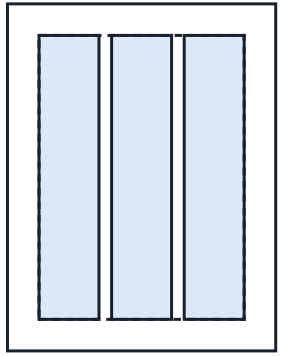
MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q1/5

3520

QUESTION ID
S105-M-Q1

TYPE
Short answer

CORRECT ANSWER
2.03 in.

GRADING NOTE Rounding to 0.01 required. (6.5 − 0.4) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q1/5

3520

ANSWER KEY Reduced copy of the question page, with its answer

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

A. 120 cm

B. 97.1 cm

C. 37.1 cm

D. 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q2/5

3521

QUESTION ID
S105-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. 97.1 cm

GRADING NOTE 60 × 1.618.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q3/5

3522

QUESTION ID

S105-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.427 in.

GRADING NOTE Rounding to 0.001 required. (6.5 − 1.375) ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A web page 1,200 px wide uses a 12-column grid with 24 px gutters between columns and no outer margins. Find the column width and the width of an element that spans 4 columns (including the gutters inside it).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q4/5

3523

QUESTION ID

S105-M-Q4

TYPE

Short answer

CORRECT ANSWER

Column 78 px; 4-column element 384 px

GRADING NOTE Form required: both. (1,200 − 11 · 24) ÷ 12 = 78; 4 · 78 + 3 · 24 = 384.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 in. by 9 in. page has 0.75 in. margins on all sides. A modular grid has 3 columns and 4 rows with 0.25 in. gutters in both directions. Find the width and height of each module to the nearest hundredth of an inch.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q5/5

3524

QUESTION ID

S105-M-Q5

TYPE

Short answer

CORRECT ANSWER

Width 1.33 in.; height 1.69 in.

GRADING NOTE Form required: both, rounded to 0.01. Text area 4.5 by 7.5; (4.5 − 2 · 0.25) ÷ 3 and (7.5 − 3 · 0.25) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 1

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q1/5

3526

QUESTION ID

S105-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16.18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q2/5

3527

QUESTION ID

S105-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.03 in.

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

A. 60 cm tall

B. 20 cm tall

C. 33 cm tall

D. 45 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q3/5

3528

QUESTION ID

S105-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 45 cm tall

GRADING NOTE 15 per part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q3/5

True or false?

Equal columns divide the available width equally.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q4/5

3529

QUESTION ID

S105-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 : 2 ratio means the second is half the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q5/5

3530

QUESTION ID

S105-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 2

Operate with decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$8.5 - 2 \times 1 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q1/5

3532

QUESTION ID

S105-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$11 \times 0.125 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q2/5

3533

QUESTION ID

S105-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.375

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

6.1 ÷ 3 to the nearest hundredth is

A. 2.3

B. 2.04

C. 2.03

D. 18.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q3/5

3534

QUESTION ID

S105-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.03

GRADING NOTE 2.0333....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q3/5

3534

True or false?

$5.125 \div 12 \approx 0.427$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q4/5

3535

QUESTION ID

S105-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$0.2 + 0.2 = 0.04$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q5/5

3536

QUESTION ID

S105-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 3

Factors of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

ANSWER KEY Reduced copy of the question page, with its answer

List all factors of 12.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q1/5

3538

QUESTION ID

S105-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 4, 6, 12

GRADING NOTE Form required: all six.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q1/5

3538

ANSWER KEY Reduced copy of the question page, with its answer

Which numbers from 2 to 6 are factors of 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q2/5

3539

QUESTION ID

S105-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2 and 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a factor of 12?

A. 8

B. 4

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q3/5

3540

QUESTION ID

S105-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 12 ÷ 4 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q3/5

3540

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

12 has more factors than 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q4/5

3541

QUESTION ID

S105-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Six factors vs four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5 is a factor of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q5/5

3542

QUESTION ID

S105-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 12 ÷ 5 is not whole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q5/5

3542

SET 105 • HSG-MG.A.3

FORWARD 1

Choose grids that support several layouts • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A 12-column grid can also be used as

A. 4 columns of 3, 3 of 4, or 2 of 6

B. 5 equal columns

C. 7 equal columns

D. only 12 columns

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q1/5

3544

QUESTION ID

S105-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 4 columns of 3, 3 of 4, or 2 of 6

GRADING NOTE Divisors of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q1/5

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q2/5

3545

QUESTION ID

S105-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 divides evenly into 2, 3, 4, and 6, giving many layout options.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q2/5

True or false?

A 10-column grid can be split into 3 equal groups of whole columns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q3/5

3546

QUESTION ID

S105-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is not a factor of 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid has 78 px columns and 24 px gutters. How wide is an element that spans 6 columns?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q4/5

3547

QUESTION ID

S105-F1-Q4

TYPE

Short answer

CORRECT ANSWER

588 px

GRADING NOTE Any equivalent form. 6 · 78 + 5 · 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6.5 in. text area must hold 4 columns, each 1.5 in. wide, with equal gutters between them. Find the exact gutter width.

Gutter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q5/5

3548

QUESTION ID
S105-F1-Q5

TYPE
Short answer

CORRECT ANSWER
⅙ in. ≈ 0.17 in.

GRADING NOTE Exact value or rounding to 0.01. (6.5 − 6) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

FORWARD 2

Scale a grid layout to a new page size • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A poster layout is 60 cm wide and 97.1 cm tall. It is reduced to a flyer 24 cm wide with the same shape. Find the scale factor and the flyer's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q1/5

3550

QUESTION ID

S105-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Factor 0.4; height 38.84 cm

GRADING NOTE Form required: both. 24 ÷ 60; 97.1 × 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the poster above, a column is 15 cm wide and a gutter is 1.2 cm. Find their widths on the flyer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q2/5

3551

QUESTION ID

S105-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Column 6 cm; gutter 0.48 cm

GRADING NOTE Form required: both. Multiply by 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q2/5

True or false?

Scaling a layout by factor k multiplies the area of each module by k^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q3/5

3552

QUESTION ID

S105-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar figures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A layout is enlarged by factor 1.5. A 4 cm by 6 cm module becomes

A. 5.5 cm by 7.5 cm

B. 4 cm by 9 cm

C. 9 cm by 13.5 cm

D. 6 cm by 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q4/5

3553

QUESTION ID

S105-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6 cm by 9 cm

GRADING NOTE Multiply both by 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a grid keeps its proportions when every measurement is multiplied by the same factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q5/5

3554

QUESTION ID

S105-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Every ratio of two measurements is unchanged (k·a/k·b = a/b), so the scaled layout is similar to the original.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

Derive and use the formula A = ½ab sin C

MAIN	Geometry main question	**HSG-SRT.D.9**
BACKWARD 1	The sine ratio in a right triangle	**HSG-SRT.C.6**
BACKWARD 2	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 3	The included angle of two sides	**HSG-CO.B.8**
FORWARD 1	Solve for a side or an angle from an area	**HSG-SRT.D.9**
FORWARD 2	From equal areas to the Law of Sines	**HSG-SRT.D.10**

SET 88 • HSG-SRT.D.9

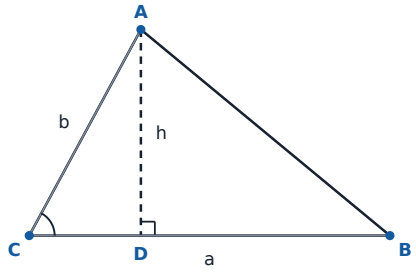
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A meets line CB at D and has length h. Derive the formula Area = ½ab sin C. Your derivation must cover both cases: ∠C acute (D on side CB, as shown) and ∠C obtuse (D on line CB extended beyond C). (If ∠C is right, h = b and sin C = 1.)

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

3557

QUESTION ID

S88-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Acute: h = b sin C. Obtuse: h = b sin(180° − C) = b sin C. In both cases Area = ½ah = ½ab sin C.

GRADING NOTE REQUIRED WORK: Express h using sine in right triangle ADC (acute case) or using the angle 180° − C (obtuse case), then use Area = ½ · base · height.

Any valid proof (paragraph, two-column, or flow). Essential: acute case sin C = h/b in right triangle ADC; obtuse case the right triangle ADC has angle 180° − C at C, so h = b sin(180° − C) and sin(180° − C) = sin C; Area = ½ · a · h in both cases. A derivation that covers only the acute case earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

3558

QUESTION ID

S88-M-Q2

TYPE

Short answer

CORRECT ANSWER

28.3

GRADING NOTE Rounding to 0.1 required. ½ · 8 · 11 · sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

3559

QUESTION ID

S88-M-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. sin 150° = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

3560

QUESTION ID

S88-M-Q4

TYPE

Short answer

CORRECT ANSWER

68.8 cm²

GRADING NOTE Rounding to 0.1 required. Two congruent triangles: 7 · 12 · sin 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

Triangle PQR has PQ = 9, QR = 14 and an obtuse angle Q = 115°. Find its area to the nearest tenth, and explain why ½ab sin C still works when the included angle is obtuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

3561

QUESTION ID

S88-M-Q5

TYPE

Short answer

CORRECT ANSWER

57.1; the altitude falls outside the triangle and has length 9 sin(180° − 115°) = 9 sin 115°, since sin(180° − Q) = sin Q.

GRADING NOTE Form required: the area (rounding to 0.1) and the explanation. ½ · 9 · 14 · sin 115°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

3563

QUESTION ID

S88-B1-Q1

TYPE

Short answer

CORRECT ANSWER

opposite ÷ hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

3564

QUESTION ID

S88-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin C = h/b, then h =

A. b + sin C

B. sin C/b

C. b/sin C

D. b sin C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

3565

QUESTION ID

S88-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b sin C

GRADING NOTE Multiply by b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 90° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

3566

QUESTION ID

S88-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin C = adjacent ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

3567

QUESTION ID

S88-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is cos C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

3569

QUESTION ID

S88-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

True or false?

Area = ½ · base · height

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

3570

QUESTION ID

S88-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

The height must be ___ to the base.

A. equal

B. adjacent

C. parallel

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

3571

QUESTION ID

S88-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. perpendicular

GRADING NOTE Altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

3572

QUESTION ID

S88-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ab sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

True or false?

A slanted side can be used as the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

3573

QUESTION ID

S88-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

3575

QUESTION ID

S88-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠A

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

True or false?

SAS uses the angle between the two given sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

3576

QUESTION ID

S88-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For sides a (BC) and b (CA), the included angle is

A. C

B. none

C. A

D. B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

3577

QUESTION ID

S88-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C

GRADING NOTE Shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

3578

QUESTION ID

S88-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The altitude is built from side b and the angle between b and the base a.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any angle of the triangle works in ½ab sin C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

3579

QUESTION ID

S88-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the included one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

3581

QUESTION ID
S88-F1-Q1

TYPE
Short answer

CORRECT ANSWER
30° or 150°

GRADING NOTE Form required: both. sin C = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

3582

QUESTION ID

S88-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.1

GRADING NOTE Rounding to 0.1 required. 50 = ½ · 20 · b · sin 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

True or false?

Two different triangles can have the same two sides and the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

3583

QUESTION ID

S88-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE C and 180° − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

A. 60°

B. 180°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

3584

QUESTION ID

S88-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 90°

GRADING NOTE sin C = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

3585

QUESTION ID
S88-F1-Q5

TYPE
Short answer

CORRECT ANSWER
9√3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

3587

QUESTION ID

S88-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

½bc sin A = ½ac sin B = ½ab sin C; dividing by ½abc gives sin A/a = sin B/b = sin C/c.

GRADING NOTE REQUIRED WORK: Set the three expressions equal and divide by ½abc.

Any valid proof (paragraph, two-column, or flow). Essential: three area expressions equal; division by ½abc; result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

3588

QUESTION ID

S88-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can be derived from the area formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

3589

QUESTION ID
S88-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE As above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

A. a sin A = b sin B

B. sin A/a = sin B/b

C. sin A = sin B

D. a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

3590

QUESTION ID

S88-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. sin A/a = sin B/b

GRADING NOTE Cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

3591

QUESTION ID

S88-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Side lengths are positive, so ½abc ≠ 0.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Sines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	Use trigonometry in right triangles	**HSG-SRT.C.8**
BACKWARD 2	The area formula ½ab sin C	**HSG-SRT.D.9**
BACKWARD 3	Proportions and cross-multiplying	**7.RP.A.2**
FORWARD 1	Use the Law of Sines (AAS and ASA)	**HSG-SRT.D.10**
FORWARD 2	Surveying with the Law of Sines	**HSG-SRT.D.11**

SET 89 • HSG-SRT.D.10

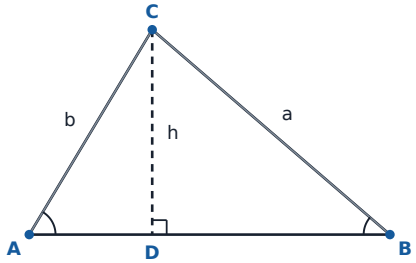
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

3594

QUESTION ID

S89-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

h = b sin A and h = a sin B, so b sin A = a sin B and a/sin A = b/sin B.

GRADING NOTE REQUIRED WORK: Write h two ways, using right triangles ADC and BDC.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = h/b and sin B = h/a from the two right triangles; equate h; divide by sin A sin B. Repeating with another altitude gives c/sin C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

3595

QUESTION ID

S89-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

h = b sin(180° − A) = b sin A and h = a sin B, so a/sin A = b/sin B.

GRADING NOTE

REQUIRED WORK: Use the right triangle with angle 180° − A at A, and sin(180° − A) = sin A.

Any valid proof (paragraph, two-column, or flow). Essential: the exterior angle at A is 180° − A; h = b sin(180° − A); the identity sin(180° − A) = sin A; equate with h = a sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, AE is the altitude from A to BC. Prove that b/sin B = c/sin C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

3596

QUESTION ID

S89-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AE = c sin B and AE = b sin C, so c sin B = b sin C and b/sin B = c/sin C.

GRADING NOTE

REQUIRED WORK: Write the altitude AE two ways, using right triangles ABE and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: sin B = AE/c and sin C = AE/b from the two right triangles; equate; divide by sin B sin C. (If ∠B or ∠C is obtuse, use sin(180° − x) = sin x.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has a right angle at C. Prove that a/sin A = b/sin B = c/sin C in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

3597

QUESTION ID

S89-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

sin A = a/c and sin B = b/c, so a/sin A = c = b/sin B; and c/sin C = c/1 = c.

GRADING NOTE

REQUIRED WORK: Use the right-triangle definitions of sine and sin 90° = 1.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = a/c, sin B = b/c (opposite over hypotenuse); each ratio equals c; sin 90° = 1 gives c/sin C = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC with acute angle A is inscribed in a circle of radius R. Draw the diameter BD. Prove that a/sin A = 2R, and explain why this gives the Law of Sines.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3598

QUESTION ID

S89-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

∠BDC ≅ ∠A (inscribed angles on arc BC); ∠BCD = 90°; sin D = BC/BD = a/(2R), so a/sin A = 2R. The same holds for b and c, so all three ratios equal 2R.

GRADING NOTE REQUIRED WORK: Compare ∠A with ∠D (same arc BC) and use right triangle BCD (angle in a semicircle).

Any valid proof (paragraph, two-column, or flow). Essential: inscribed angles on the same arc; angle in a semicircle; sine in right triangle BCD; each ratio equals 2R. (For obtuse A, ∠D = 180° − A and sin D = sin A.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3598

SET 89 • HSG-SRT.D.10

BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

3600

QUESTION ID

S89-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

True or false?

In an acute triangle, an altitude splits the triangle into two right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

3601

QUESTION ID

S89-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The foot lies on the opposite side; in an obtuse triangle, an altitude from an acute-angle vertex falls outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin B = h/a, then h =

A. h sin B

B. a/sin B

C. a sin B

D. sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

3602

QUESTION ID

S89-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a sin B

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

3603

QUESTION ID
S89-B1-Q4

TYPE
Short answer

CORRECT ANSWER
30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

True or false?

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

3604

QUESTION ID
S89-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The altitude creates right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

3606

QUESTION ID

S89-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

True or false?

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

3607

QUESTION ID

S89-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with sides 6, 6 and included 30°?

A. 3

B. 9

C. 36

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

3608

GRADING NOTE ½ · 36 · ½.

QUESTION ID

S89-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

3609

QUESTION ID

S89-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ac sin B

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

½ab sin C uses angle A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

3610

QUESTION ID

S89-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

3612

QUESTION ID

S89-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

True or false?

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

3613

QUESTION ID

S89-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by sin A sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From 3x = 4y, x/y =

A. 3/4

B. 4/3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

3614

QUESTION ID

S89-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/3

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite p · q = r · s as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

3615

QUESTION ID

S89-B3-Q4

TYPE

Short answer

CORRECT ANSWER

p/r = s/q (or p/s = r/q)

GRADING NOTE Any correct proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

True or false?

a/b = c/d means a + d = b + c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

3616

QUESTION ID

S89-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

3618

QUESTION ID

S89-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13.5

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

3619

QUESTION ID

S89-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C = 80°; c ≈ 15.3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

True or false?

Knowing two angles and any side determines the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

3620

QUESTION ID

S89-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AAS/ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 45°, a = 6. b =

A. 3√2 ≈ 4.2

B. 6√2 ≈ 8.5

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

3621

QUESTION ID

S89-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6√2 ≈ 8.5

GRADING NOTE 6 sin 45°/sin 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

3622

QUESTION ID

S89-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The law pairs each side with its opposite angle, so C is needed for c.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

3624

QUESTION ID

S89-F2-Q1

TYPE

Short answer

CORRECT ANSWER

565 m

GRADING NOTE Rounding to the nearest meter required. Third angle 50°; side opposite 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

3625

QUESTION ID
S89-F2-Q2

TYPE
Short answer

CORRECT ANSWER
225 m

GRADING NOTE Rounding to the nearest meter required. Third angle 57°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

True or false?

Surveyors use a measured baseline and two angles to find distances they cannot walk.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

3626

QUESTION ID

S89-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Triangulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

A. equal to the others

B. the shortest

C. the longest

D. the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

3627

QUESTION ID

S89-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the longest

GRADING NOTE Law of Sines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

3628

QUESTION ID
S89-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The far side cannot be reached; angles can be measured from one bank.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Cosines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	The distance formula	**8.G.B.8**
BACKWARD 3	Sine and cosine in a right triangle	**HSG-SRT.C.6**
BACKWARD 4	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Use the Law of Cosines (SAS and SSS)	**HSG-SRT.D.10**
FORWARD 2	Distances between moving objects	**HSG-SRT.D.11**

SET 90 • HSG-SRT.D.10

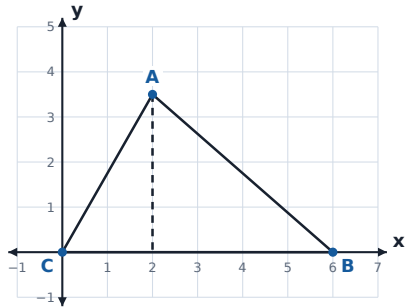
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

3631

QUESTION ID

S90-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$c^2 = (b \cos C - a)^2 + (b \sin C)^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE REQUIRED WORK: Use the distance formula for AB and the identity $\sin^2 C + \cos^2 C = 1$.

Any valid proof (paragraph, two-column, or flow). Essential: A = (b cos C, b sin C) because CA = b; $c^2 = (b \cos C - a)^2 + b^2 \sin^2 C$; expand: $b^2\cos^2 C - 2ab \cos C + a^2 + b^2 \sin^2 C$; $\sin^2 C + \cos^2 C = 1$ gives $a^2 + b^2 - 2ab \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

3632

QUESTION ID

S90-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$h^2 = b^2 - x^2 = c^2 - (a - x)^2$, so $c^2 = a^2 + b^2 - 2ax = a^2 + b^2 - 2ab \cos C$.

GRADING NOTE REQUIRED WORK: Write h^2 twice with the Pythagorean Theorem, then use $x = b \cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $BD = a - x$; $h^2 = b^2 - x^2$ and $h^2 = c^2 - (a - x)^2$; solve for c^2; substitute $x = b \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, ∠C is obtuse, so the altitude from A meets line BC extended beyond C at D. Let CD = x. Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

3633

QUESTION ID

S90-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$h^2 = b^2 - x^2 = c^2 - (a + x)^2$, so $c^2 = a^2 + b^2 + 2ax = a^2 + b^2 - 2ab \cos C$.

GRADING NOTE REQUIRED WORK: Write h^2 twice with the Pythagorean Theorem, using BD = a + x, then use $x = b \cos(180° - C) = -b \cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: BD = a + x; two expressions for h^2; solve for c^2; $x = b \cos(180° - C)$ and $\cos(180° - C) = -\cos C$, so $2ax = -2ab \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Law of Cosines to prove that in triangle ABC, if $c^2 > a^2 + b^2$, then $\angle C$ is obtuse.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

3634

QUESTION ID

S90-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

cos C = (a² + b² − c²)/(2ab) < 0, so C is between 90° and 180°: obtuse.

GRADING NOTE

REQUIRED WORK: Solve the law for cos C and decide its sign.

Any valid proof (paragraph, two-column, or flow). Essential: rearrangement; numerator negative and denominator positive; a negative cosine means an obtuse angle ($0° < C < 180°$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Law of Cosines to prove both directions: in triangle ABC, $\angle C = 90°$ if and only if $c^2 = a^2 + b^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

3635

QUESTION ID

S90-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$C = 90°$ gives $c^2 = a^2 + b^2$; $c^2 = a^2 + b^2$ gives $\cos C = 0$, so $C = 90°$.

GRADING NOTE REQUIRED WORK: Substitute $\cos 90° = 0$ for one direction; for the converse solve for $\cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $\cos 90° = 0$; converse: $\cos C = 0$ with $0° < C < 180°$ forces $C = 90°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

3637

QUESTION ID

S90-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

3638

QUESTION ID

S90-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

A. a^2

B. c^2

C. x^2

D. b^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

3639

QUESTION ID

S90-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c^2

GRADING NOTE Right triangle ABD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

True or false?

The Pythagorean Theorem applies to the right triangles made by an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

3640

QUESTION ID

S90-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angles at D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

3640

True or false?

$a^2 + b^2 = c^2$ in every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

3641

QUESTION ID

S90-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

3643

QUESTION ID

S90-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

3644

QUESTION ID

S90-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(p - a)^2 + q^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

Squared distance from (1, 2) to (4, 6)?

A. 5

B. 25

C. 49

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

3645

QUESTION ID

S90-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

3646

QUESTION ID

S90-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

True or false?

Distance from (2, 0) to (5, 0) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

3647

QUESTION ID

S90-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

3649

QUESTION ID

S90-B3-Q1

TYPE

Short answer

CORRECT ANSWER

b cos C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

3650

QUESTION ID

S90-B3-Q2

TYPE

Short answer

CORRECT ANSWER

b sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

A. 25

B. 1

C. 12/25

D. 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

3651

QUESTION ID

S90-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE 9/25 + 16/25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

3652

QUESTION ID

S90-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

cos C = opposite ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

3653

QUESTION ID
S90-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Adjacent ÷ hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

3655

QUESTION ID
S90-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$b^2\cos^2 C - 2ab\cos C + a^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

3656

QUESTION ID
S90-B4-Q2

TYPE
Short answer

CORRECT ANSWER
$a^2 - 2ax + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(p - q)^2 =$

A. $p^2 - 2pq + q^2$

B. $p^2 - pq + q^2$

C. $p^2 - q^2$

D. $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

3657

QUESTION ID
S90-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $p^2 - 2pq + q^2$

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 - x^2 = a^2 - 2ax$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

3658

QUESTION ID
S90-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE x^2 cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 = a^2 - x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

3659

QUESTION ID
S90-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing −2ax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

3661

QUESTION ID
S90-F1-Q1

TYPE
Short answer

CORRECT ANSWER
7.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

3662

QUESTION ID

S90-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. cos = (25 + 64 − 49)/80 = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

True or false?

The Law of Cosines can find an angle when all three sides are known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

3663

QUESTION ID

S90-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

Sides 3, 4, 6. The angle opposite 6 is

A. obtuse

B. acute

C. right

D. impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

3664

QUESTION ID

S90-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. obtuse

GRADING NOTE 36 > 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

3665

QUESTION ID
S90-F1-Q5

TYPE
Short answer

CORRECT ANSWER
82.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

3667

QUESTION ID

S90-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29.8 km

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

3668

QUESTION ID
S90-F2-Q2

TYPE
Short answer

CORRECT ANSWER
46 yd

GRADING NOTE Rounding to the nearest yard required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

3669

QUESTION ID

S90-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It works for any included angle; with 90° it becomes the Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

A. 8 km

B. √34 ≈ 5.8 km

C. 2 km

D. √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

3670

QUESTION ID

S90-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √19 ≈ 4.4 km

GRADING NOTE 9 + 25 − 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

3670

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

3671

QUESTION ID

S90-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The angle between their courses is not 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Sines, including the ambiguous case

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Sines	**HSG-SRT.D.10**
BACKWARD 2	Inverse sine	**HSG-SRT.C.8**
BACKWARD 3	Triangle angle sum	**HSG-CO.C.10**
BACKWARD 4	Sine of supplementary angles	**HSG-SRT.D.9**
FORWARD 1	Solve both triangles in the ambiguous case	**HSG-SRT.D.11**
FORWARD 2	Bearings and navigation with the Law of Sines	**HSG-MG.A.1**

SET 91 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. ∠A = 72° and ∠B = 63°. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

3674

QUESTION ID

S91-M-Q1

TYPE

Short answer

CORRECT ANSWER

151.2 m

GRADING NOTE Rounding to 0.1 required. ∠C = 45°; AC is opposite ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

3675

QUESTION ID

S91-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two: B ≈ 56.4° or B ≈ 123.6°

GRADING NOTE Form required: the number and both angles. sin B = 10 sin 30°/6 ≈ 0.833.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

3676

QUESTION ID

S91-M-Q3

TYPE

Short answer

CORRECT ANSWER

One: B ≈ 32.4°

GRADING NOTE Form required: the number and the angle. The supplement 147.6° is too large (with A = 40° the sum exceeds 180°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

3677

QUESTION ID

S91-M-Q4

TYPE

Short answer

CORRECT ANSWER

None: sin B = 10 sin 50°/5 ≈ 1.53 > 1.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

A is acute and b sin A < a < b. How many triangles have these measurements?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

3678

QUESTION ID

S91-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Two

GRADING NOTE Side a can swing to two positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

3680

QUESTION ID

S91-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a/sin A = b/sin B = c/sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

3681

QUESTION ID

S91-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

Solving a/sin A = b/sin B for sin B gives

A. b sin A/a

B. a/(b sin A)

C. sin A

D. a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

3682

QUESTION ID

S91-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b sin A/a

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

True or false?

Each side in the Law of Sines is paired with its opposite angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

3683

QUESTION ID

S91-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a/\sin B = b/\sin A$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

3684

QUESTION ID

S91-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Pairs are opposite.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

3686

QUESTION ID

S91-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

3687

QUESTION ID

S91-B2-Q2

TYPE

Short answer

CORRECT ANSWER

53.1°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin^{-1}(1) =$

A. 0°

B. 180°

C. 45°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

3688

QUESTION ID

S91-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90°

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

3689

QUESTION ID

S91-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(1.2)$ is an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

3690

QUESTION ID

S91-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sine cannot exceed 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

3692

QUESTION ID

S91-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

True or false?

A triangle cannot have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

3693

QUESTION ID

S91-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

A = 40° and B = 147.6°. Is this a triangle?

A. Only if C is 0°

B. No: the sum exceeds 180°

C. Yes

D. Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

3694

QUESTION ID

S91-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No: the sum exceeds 180°

GRADING NOTE 187.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

3695

QUESTION ID

S91-B3-Q4

TYPE

Short answer

CORRECT ANSWER

26.4°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angles of a triangle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

3696

QUESTION ID

S91-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 150° = \sin 30°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

3698

QUESTION ID

S91-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sin(180° - x) = \sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

3699

QUESTION ID

S91-B4-Q2

TYPE

Short answer

CORRECT ANSWER

123.6°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 120° equals

A. cos 120°

B. sin 60°

C. sin 30°

D. −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

3700

QUESTION ID

S91-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. sin 60°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

3701

QUESTION ID

S91-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Both B and 180° − B have that sine.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An obtuse angle has a negative sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

3702

QUESTION ID

S91-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

3704

QUESTION ID

S91-F1-Q1

TYPE

Short answer

CORRECT ANSWER

C ≈ 93.6°; c ≈ 12.0

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

3705

QUESTION ID

S91-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C ≈ 26.4°; c ≈ 5.3

GRADING NOTE Form required: both. B ≈ 123.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

True or false?

In the ambiguous case the two triangles have different third sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

3706

QUESTION ID
S91-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

A = 30°, b = 10, and a = 5. How many triangles?

A. Three

B. Two

C. One (a right triangle)

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

3707

QUESTION ID

S91-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. One (a right triangle)

GRADING NOTE a = b sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 120°, a = 8, b = 10. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

3708

QUESTION ID

S91-F1-Q5

TYPE

Short answer

CORRECT ANSWER

None: with A obtuse, a must be the longest side, but a < b.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3710

QUESTION ID

S91-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8.0 km

GRADING NOTE Rounding to 0.1 required. The triangle has angles 35°, 110°, 35°, so it is isosceles: 8.0 km.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3710

ANSWER KEY Reduced copy of the question page, with its answer

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

3711

QUESTION ID

S91-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9.3 km

GRADING NOTE Rounding to 0.1 required. Third angle 71°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

True or false?

Bearings and sightings are modeled as triangles with a known side and two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

3712

QUESTION ID

S91-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

A. 75°

B. 110°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

3713

QUESTION ID

S91-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 70° − 35° (exterior angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

3713

ANSWER KEY Reduced copy of the question page, with its answer

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

3714

QUESTION ID

S91-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Earth's surface is curved.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Cosines, including resultant forces

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Cosines	**HSG-SRT.D.10**
BACKWARD 2	Inverse cosine	**HSG-SRT.C.8**
BACKWARD 3	Angles in a parallelogram	**HSG-CO.C.11**
FORWARD 1	Solve a triangle with both laws	**HSG-SRT.D.11**
FORWARD 2	Velocity vectors: airspeed and wind	**HSN-VM.A.3**

SET 92 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

3717

QUESTION ID

S92-M-Q1

TYPE

Short answer

CORRECT ANSWER

434.5 m

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

3718

QUESTION ID

S92-M-Q2

TYPE

Short answer

CORRECT ANSWER

91.0 N

GRADING NOTE Rounding to 0.1 required. In the force parallelogram the angle opposite the resultant is 130°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

3719

QUESTION ID
S92-M-Q3

TYPE
Short answer

CORRECT ANSWER
91.8°

GRADING NOTE Rounding to 0.1 required. Opposite 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

3720

QUESTION ID

S92-M-Q4

TYPE

Short answer

CORRECT ANSWER

341.9 km

GRADING NOTE Rounding to 0.1 required. The angle between the legs inside the triangle is 155°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

Which law should be used first to solve a triangle given all three sides?

A. The Law of Sines

B. The Pythagorean Theorem

C. The angle sum

D. The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

3721

QUESTION ID

S92-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The Law of Cosines

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

3723

QUESTION ID

S92-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

3724

QUESTION ID

S92-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solved for cos C:

A. $(a^2 + b^2 - c^2)/(2ab)$

B. $(c^2 - a^2 - b^2)/(2ab)$

C. $2ab/(a^2 + b^2 - c^2)$

D. $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

3725

QUESTION ID

S92-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

$(a^2 + b^2 - c^2)/(2ab)$

GRADING NOTE Rearrange.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

True or false?

The angle in the formula is the one opposite the side being found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

3726

QUESTION ID

S92-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE C opposite c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$$c^2 = a^2 + b^2 + 2ab \cos C$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

3727

QUESTION ID

S92-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Minus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

cos C = 0.5. C =

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

3729

QUESTION ID

S92-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos C = −0.03125. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

3730

QUESTION ID

S92-B2-Q2

TYPE

Short answer

CORRECT ANSWER

91.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A negative cosine means the angle is

A. acute

B. right

C. obtuse

D. zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

3731

QUESTION ID

S92-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

True or false?

$\cos^{-1}$ gives angles from 0° to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

3732

QUESTION ID

S92-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0) = 0°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

3733

QUESTION ID
S92-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

3735

QUESTION ID

S92-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

3736

QUESTION ID

S92-B3-Q2

TYPE

Short answer

CORRECT ANSWER

130°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

A. 130°

B. 25°

C. 50°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

3737

QUESTION ID

S92-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 130°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

3738

QUESTION ID

S92-B3-Q4

TYPE

Short answer

CORRECT ANSWER

155°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

True or false?

The angle between two forces is the angle used in the Law of Cosines for the resultant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3739

QUESTION ID

S92-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use its supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3739

SET 92 • HSG-SRT.D.11

FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

3741

QUESTION ID

S92-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

3742

QUESTION ID

S92-F1-Q2

TYPE

Short answer

CORRECT ANSWER

53.0°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

True or false?

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

3743

QUESTION ID

S92-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Avoids the ambiguous obtuse case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

A. $\sin^{-1}$

B. the Law of Sines

C. the Law of Cosines

D. the angle sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

3744

QUESTION ID

S92-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the Law of Cosines

GRADING NOTE SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

3745

QUESTION ID

S92-F1-Q5

TYPE

Short answer

CORRECT ANSWER

40.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3747

QUESTION ID

S92-F2-Q1

TYPE

Short answer

CORRECT ANSWER

304.1 km/h at 9.5° north of east

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3747

ANSWER KEY Reduced copy of the question page, with its answer

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

3748

QUESTION ID
S92-F2-Q2

TYPE
Short answer

CORRECT ANSWER
8.5 km/h

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

True or false?

The resultant of two velocities is always found by adding their speeds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

3749

QUESTION ID

S92-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when they point the same way; otherwise use the parallelogram rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Velocities of 10 and 10 with 120° between them have a resultant of

A. 0

B. 17.3

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3750

QUESTION ID

S92-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Law of Cosines with 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3750

ANSWER KEY Reduced copy of the question page, with its answer

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

3751

QUESTION ID

S92-F2-Q5

TYPE

Short answer

CORRECT ANSWER

444.5 km/h

GRADING NOTE Rounding to 0.1 required. Angle in the triangle 135°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

Construct a tangent line from a point outside a circle

MAIN	Geometry main question	**HSG-C.A.4**
BACKWARD 1	Construct the midpoint of a segment	**HSG-CO.D.12**
BACKWARD 2	Inscribed angles on a diameter	**HSG-C.A.2**
BACKWARD 3	Tangent is perpendicular to the radius	**HSG-C.A.2**
FORWARD 1	Prove that the two tangent segments are congruent and compute…	**HSG-C.A.4**
FORWARD 2	Lines of sight to the horizon	**HSG-MG.A.1**

SET 93 • HSG-C.A.4

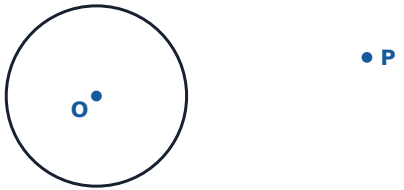
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

3754

QUESTION ID

S93-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw OP; construct its midpoint M (perpendicular bisector); draw the circle with center M through O; it meets circle O at T_1 and T_2; draw PT_1 (and PT_2).

GRADING NOTE Grade the construction marks, not only the final figure: segment OP, the perpendicular bisector arcs locating M, the circle on diameter OP, and the tangent through a crossing point T.

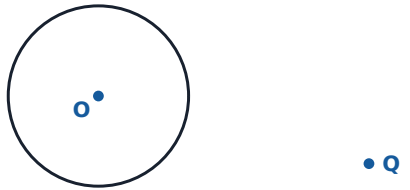
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

3755

QUESTION ID

S93-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same construction; both intersection points T_1 and T_2 give the two tangents QT_1 and QT_2.

GRADING NOTE Grade the construction marks, not only the final figure: all construction marks and both tangents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through R tangent to the circle with center O. Then write one sentence explaining why your line is tangent.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

3756

QUESTION ID

S93-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Midpoint M of OR; circle with center M through O meets circle O at T; line RT. Tangent because ∠OTR is inscribed in a semicircle (90°), so RT ⊥ radius OT at T.

GRADING NOTE Grade the construction marks, not only the final figure: all construction marks for the midpoint and the helper circle, the tangent line, and the right-angle justification.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct both tangent lines from S to the circle with center O. Then, as evidence, set your compass to ST_1 and show that the same opening reaches T_2 from S.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

3757

QUESTION ID

S93-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Construct both tangents ST_1 and ST_2 with the circle on diameter OS; the compass check shows $ST_1 = ST_2$ (congruent tangent segments).

GRADING NOTE Grade the construction marks, not only the final figure: both tangents from the full construction and the compass check of equal tangent segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct both tangent lines from P to the circle with center O. Then construct the bisector of the angle formed by the two tangents and show that it passes through O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

3758

QUESTION ID

S93-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Both tangents from the circle on diameter OP; bisecting $\angle T_1PT_2$ gives a ray through O (OP bisects the angle between the tangents).

GRADING NOTE Grade the construction marks, not only the final figure: both tangents, the angle bisector construction, and the observation that it passes through O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

3760

QUESTION ID

S93-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of OP

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector crosses the segment at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

3761

QUESTION ID

S93-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The perpendicular bisector construction uses arcs

A. of any radius

B. centered at the midpoint

C. of equal radius from both endpoints

D. from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

3762

QUESTION ID

S93-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. of equal radius from both endpoints

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

3763

QUESTION ID

S93-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It is the center of the circle with diameter OP.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The midpoint can be found by eye in a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

3764

QUESTION ID

S93-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It must be constructed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

An angle inscribed in a semicircle is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

3766

QUESTION ID

S93-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

3767

QUESTION ID

S93-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

The intercepted arc of an angle inscribed in a semicircle is

A. 180°

B. 360°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

3768

QUESTION ID

S93-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 180°

GRADING NOTE Half of 180 is 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

3769

QUESTION ID

S93-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Points on it make a right angle with O and P.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

True or false?

An inscribed angle on a diameter can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

3770

QUESTION ID

S93-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

3772

QUESTION ID

S93-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

3773

QUESTION ID

S93-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT ⊥ OT and T is on circle O, then PT is

A. a secant

B. a diameter

C. a chord

D. tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

3774

QUESTION ID

S93-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. tangent to the circle

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

3775

QUESTION ID

S93-B3-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

True or false?

A tangent passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

3776

QUESTION ID

S93-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches at one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

3778

QUESTION ID

S93-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△OAP ≅ △OBP by HL, so PA = PB.

GRADING NOTE REQUIRED WORK: Use the right triangles OAP and OBP.

Any valid proof (paragraph, two-column, or flow). Essential: right angles at A and B (tangent ⊥ radius); OA = OB (radii); OP common hypotenuse; HL; CPCTC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

3779

QUESTION ID

S93-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

True or false?

OP bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

3780

QUESTION ID

S93-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P make a 60° angle. The angle AOB is

A. 30°

B. 120°

C. 60°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

3781

QUESTION ID

S93-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 120°

GRADING NOTE Quadrilateral with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

3782

QUESTION ID

S93-F1-Q5

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction (light travels in straight lines). A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

3784

QUESTION ID

S93-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5.0 km

GRADING NOTE Rounding to 0.1 required. $d^2 = (6371.002)^2 - 6371^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction. From the top of a 100 m (0.1 km) tower at sea level, how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3785

QUESTION ID
S93-F2-Q2

TYPE
Short answer

CORRECT ANSWER
36 km

GRADING NOTE Rounding to the nearest km required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3785

True or false?

The line of sight to the horizon passes through Earth's center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3786

QUESTION ID

S93-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is tangent, perpendicular to the radius at the horizon point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3786

ANSWER KEY Reduced copy of the question page, with its answer

Raising the eye height by a factor of 4 multiplies the horizon distance by about

A. 16

B. 8

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

3787

QUESTION ID

S93-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $d \approx \sqrt{(2Rh)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

3788

QUESTION ID

S93-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The tangent line of sight is perpendicular to the radius at the point of tangency.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of an ellipse from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive a circle equation	**HSG-GPE.A.1**
BACKWARD 3	Expand squares	**HSA-APR.A.1**
FORWARD 1	Read foci and axes from an ellipse equation	**HSG-GPE.A.3**
FORWARD 2	Design with the reflective property of ellipses	**HSG-MG.A.3**

SET 94 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

3791

QUESTION ID

S94-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$$x^2/25 + y^2/16 = 1$$

GRADING NOTE REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + 3)^2 + y^2)} + \sqrt{((x - 3)^2 + y^2)} = 10$; isolate and square; simplify to $3x - 25 = -5\sqrt{((x - 3)^2 + y^2)}$ (or equivalent); square again; $16x^2 + 25y^2 = 400$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

3792

QUESTION ID

S94-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x^2/9 + y^2/25 = 1$

GRADING NOTE REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y + 4)^2)} + \sqrt{(x^2 + (y - 4)^2)} = 10$; isolate and square; $4y - 25 = -5\sqrt{(x^2 + (y - 4)^2)}$ (or equivalent); square again; $25x^2 + 9y^2 = 225$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

3793

QUESTION ID

S94-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$x^2/169 + y^2/144 = 1$

GRADING NOTE REQUIRED WORK: The vertices give the constant sum 2a = 26; then derive as before.

Any valid proof (paragraph, two-column, or flow). Essential: sum 26 (the vertex (13, 0) is 18 and 8 from the foci); $\sqrt{((x + 5)^2 + y^2)} + \sqrt{((x - 5)^2 + y^2)} = 26$; two squarings; $144x^2 + 169y^2 = 24{,}336$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10. (You may translate the figure so the center is at the origin, derive, then translate back.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

3794

QUESTION ID

S94-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$(x - 4)^2/25 + (y - 2)^2/16 = 1$

GRADING NOTE REQUIRED WORK: Find the center (4, 2); translate to foci (±3, 0); derive; replace x with x − 4 and y with y − 2.

Any valid proof (paragraph, two-column, or flow). Essential: center (4, 2); translated foci (±3, 0) with sum 10 give $x^2/25 + y^2/16 = 1$; translating back.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci $(-c, 0)$ and $(c, 0)$ whose distances add to $2a$ $(a > c > 0)$. Show that it can be written $x^2/a^2 + y^2/b^2 = 1$ with $b^2 = a^2 - c^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

3795

QUESTION ID

S94-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$x^2/a^2 + y^2/b^2 = 1$, $b^2 = a^2 - c^2$

GRADING NOTE REQUIRED WORK: Isolate one radical, square twice, and divide by $a^2(a^2 - c^2)$.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + c)^2 + y^2)} = 2a - \sqrt{((x - c)^2 + y^2)}$; square: $a\sqrt{((x - c)^2 + y^2)} = a^2 - cx$; square: $(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$; divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

3797

QUESTION ID

S94-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{((x - 3)^2 + y^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

3798

QUESTION ID

S94-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 0) to (−3, 0)?

A. √34

B. 64

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

3799

QUESTION ID

S94-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The distance formula has a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

3800

QUESTION ID

S94-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

3800

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(a^2 + b^2)} = a + b$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

3801

QUESTION ID

S94-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

3803

QUESTION ID

S94-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

True or false?

A circle is an ellipse whose two foci coincide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

3804

QUESTION ID

S94-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Special case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2/25 + y^2/25 = 1$ is

A. an ellipse with different axes

B. a parabola

C. a circle of radius 5

D. a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

3805

QUESTION ID

S94-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a circle of radius 5

GRADING NOTE Equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

3806

QUESTION ID

S94-B2-Q4

TYPE

Short answer

CORRECT ANSWER

one point (the center)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An ellipse has one focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

3807

QUESTION ID

S94-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

3809

QUESTION ID
S94-B3-Q1

TYPE
Short answer

CORRECT ANSWER
12x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

3810

QUESTION ID
S94-B3-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(3x - 25)^2 =$

A. $9x^2 + 625$

B. $9x^2 - 75x + 625$

C. $3x^2 - 150x + 625$

D. $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

3811

QUESTION ID
S94-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $9x^2 - 150x + 625$

GRADING NOTE Middle term $2 \cdot 3x \cdot 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

3812

QUESTION ID
S94-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

3813

QUESTION ID
S94-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

3815

QUESTION ID

S94-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±4, 0)

GRADING NOTE Form required: both points. $c^2 = 25 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

3816

QUESTION ID

S94-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±5); foci (0, ±3)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

True or false?

For an ellipse, $c^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

3817

QUESTION ID

S94-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

Which ellipse is closest to a circle?

A. $x^2/100 + y^2/9 = 1$

B. $x^2/25 + y^2/4 = 1$

C. $x^2/25 + y^2/24 = 1$

D. $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

3818

QUESTION ID

S94-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/25 + y^2/24 = 1$

GRADING NOTE Nearly equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

3819

QUESTION ID

S94-F1-Q5

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

3821

QUESTION ID

S94-F2-Q1

TYPE

Short answer

CORRECT ANSWER

18.3 m

GRADING NOTE Rounding to 0.1 required. a = 20, b = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

3822

QUESTION ID

S94-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At the other focus, 6 cm from the center: (−6, 0) if the stone is at (6, 0).

GRADING NOTE Any equivalent answer. c = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

True or false?

Sound from one focus of an elliptical room reflects back to the same focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

3823

QUESTION ID

S94-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It reflects to the other focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

A. 8 m

B. 10 m

C. 4 m

D. 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3824

QUESTION ID

S94-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8 m

GRADING NOTE a = 5, c = 3, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3824

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

3825

QUESTION ID

S94-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The string length is the constant sum of the distances to the two stakes (foci).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of a hyperbola from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive an ellipse equation	**HSG-GPE.A.3**
BACKWARD 3	Absolute value as distance	**6.NS.C.7**
FORWARD 1	Read foci and asymptotes from a hyperbola equation	**HSG-GPE.A.3**
FORWARD 2	Locate a position from a time difference	**HSG-MG.A.3**

SET 95 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A hyperbola is the set of points whose distances to the foci (−5, 0) and (5, 0) differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

3828

QUESTION ID

S95-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2/9 - y^2/16 = 1$

GRADING NOTE REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 6$; isolate and square; simplify to $5x - 9 = \pm 3\sqrt{((x - 5)^2 + y^2)}$ (or equivalent); square again; $16x^2 - 9y^2 = 144$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (0, ±13) whose distances to the foci differ by 10 (in absolute value).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

3829

QUESTION ID

S95-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$$\frac{y^2}{25} - \frac{x^2}{144} = 1$$

GRADING NOTE

REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{(x^2 + (y + 13)^2)} - \sqrt{(x^2 + (y - 13)^2)}| = 10$; two squarings; $144y^2 - 25x^2 = 3{,}600$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

3830

QUESTION ID

S95-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$x^2/36 - y^2/64 = 1$

GRADING NOTE REQUIRED WORK: The vertices give the constant difference 2a = 12; then derive.

Any valid proof (paragraph, two-column, or flow). Essential: difference 12 (from vertex (6, 0): 16 − 4); two squarings; $64x^2 - 36y^2 = 2{,}304$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci $(-c, 0)$ and $(c, 0)$ whose distances differ by $2a$ $(0 < a < c)$. Show that it can be written $x^2/a^2 - y^2/b^2 = 1$ with $b^2 = c^2 - a^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

3831

QUESTION ID

S95-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$x^2/a^2 - y^2/b^2 = 1$, $b^2 = c^2 - a^2$

GRADING NOTE REQUIRED WORK: Isolate one radical, square twice, and divide by $a^2(c^2 - a^2)$.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + c)^2 + y^2)} = \pm 2a + \sqrt{((x - c)^2 + y^2)}$; square: $cx - a^2 = \pm a\sqrt{((x - c)^2 + y^2)}$; square: $(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2)$; divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±5, 0) whose distances to the foci differ by 8. Then give its asymptotes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

3832

QUESTION ID

S95-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$x^2/16 - y^2/9 = 1$; asymptotes $y = \pm(3/4)x$

GRADING NOTE REQUIRED WORK: Derive as before, then read b/a.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 8$; two squarings; $9x^2 - 16y^2 = 144$; asymptotes from b/a = 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

3834

QUESTION ID

S95-B1-Q1

TYPE

Short answer

CORRECT ANSWER

√((x + 5)² + y²)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

3835

QUESTION ID

S95-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−6 (difference 6 in absolute value)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (6, 0) to (−10, 0)?

A. 4

B. 60

C. 16

D. −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

3836

QUESTION ID

S95-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

True or false?

Distances are never negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

3837

QUESTION ID

S95-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

True or false?

Distance from (0, 13) to (0, −13) is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

3838

QUESTION ID

S95-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

True or false?

An ellipse uses a constant SUM of distances to the foci.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

3840

QUESTION ID

S95-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

3841

QUESTION ID

S95-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2/25 + y^2/16 = 1$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For an ellipse, $b^2 =$

A. ac

B. $a^2 - c^2$

C. $a^2 + c^2$

D. $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

3842

QUESTION ID

S95-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a^2 - c^2$

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

3843

QUESTION ID

S95-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The difference (not the sum) of the distances is constant.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

True or false?

An ellipse equation has a minus sign between the squared terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

3844

QUESTION ID

S95-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Plus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

3846

QUESTION ID

S95-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

3847

QUESTION ID

S95-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

|a − b| means

A. the size of the difference

B. a + b

C. a negative number

D. always a − b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

3848

QUESTION ID

S95-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the size of the difference

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|x| = 6$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

3849

QUESTION ID

S95-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ±6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|-3| = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

3850

QUESTION ID

S95-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3852

QUESTION ID

S95-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±5, 0)

GRADING NOTE Form required: both points. $c^2 = 16 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3852

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

3853

QUESTION ID

S95-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±2); asymptotes y = ±(2/3)x

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

True or false?

For a hyperbola, $c^2 = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

3854

QUESTION ID

S95-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

Which equation is a hyperbola?

A. $x^2/4 + y^2/9 = 1$

B. $y = x^2/4$

C. $x^2/4 - y^2/9 = 1$

D. $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

3855

QUESTION ID

S95-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/4 - y^2/9 = 1$

GRADING NOTE Minus sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

3856

QUESTION ID

S95-F1-Q5

TYPE

Short answer

CORRECT ANSWER

An ellipse has a plus sign between the squared terms; a hyperbola has a minus sign.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

3858

QUESTION ID

S95-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2/900 - y^2/1600 = 1$

GRADING NOTE Form required: an equation. $2a = 60$, $c = 50$, $b^2 = 2500 - 900$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

3859

QUESTION ID

S95-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.02 km

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

True or false?

For one pair of stations, a single nonzero time difference (with the sign known, that is, which signal arrived first) places the ship on one branch of a hyperbola, not at a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3860

QUESTION ID
S95-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A second, independent measurement is needed to fix the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3860

ANSWER KEY Reduced copy of the question page, with its answer

Radio stations are at (−50, 0) and (50, 0) (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

A. (30, 0)

B. (40, 0)

C. (50, 0)

D. (60, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

3861

QUESTION ID

S95-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 0)

GRADING NOTE Vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

One pair of stations and one nonzero signed time difference place a ship on one branch of a hyperbola. What extra measurement is needed to fix its position, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

3862

QUESTION ID

S95-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A time difference from a second pair of stations (or another independent measurement such as a bearing); the second curve crosses the first branch, and the crossing point is the position.

GRADING NOTE Any equivalent answer naming an extra measurement and the intersection idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

Use Cavalieri's principle to justify the volume of a sphere

MAIN	Geometry main question	**HSG-GMD.A.2**
BACKWARD 1	Cavalieri's principle for prisms and cylinders	**HSG-GMD.A.1**
BACKWARD 2	Cross-section radius with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Volumes of cylinders and cones	**HSG-GMD.A.3**
BACKWARD 4	Area of a ring	**7.G.B.4**
FORWARD 1	Use Cavalieri's principle for other solids	**HSG-GMD.A.2**
FORWARD 2	Model a dome as a hemispherical shell	**HSG-MG.A.2**

SET 96 • HSG-GMD.A.2

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is ⅔πr³, and conclude that a sphere has volume 4/3 πr³.

Work space



hemisphere, radius r cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

3865

QUESTION ID

S96-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Both cross-sections have area π(r² − y²), so the volumes are equal: πr³ − ⅓πr³ = ⅔πr³; the sphere is two hemispheres: 4/3 πr³.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas at height y.

Any valid explanation. Essential: hemisphere slice radius √(r² − y²) (Pythagorean), area π(r² − y²); the other slice is a ring with outer radius r and inner radius y (the cone radius at height y), area πr² − πy²; equal areas at every height and equal heights; Cavalieri; volume cylinder minus cone = ⅔πr³; double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Let r = 6. A hemisphere of radius 6 rests on its flat base, and a cylinder of radius 6 and height 6 has a cone removed whose vertex is at the center of the bottom base and whose base is the top of the cylinder. Show that the two cross-sections have equal area at every height y ($0 \le y \le 6$), and use Cavalieri's principle to find the hemisphere's and the sphere's volumes.

Work space



hemisphere, radius r cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

3866

QUESTION ID

S96-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Hemisphere slice $\pi(36 - y^2)$; ring $\pi(36 - y^2)$; equal at every y, equal heights, so V(hemisphere) = $216\pi - 72\pi = 144\pi$ and V(sphere) = 288π.

GRADING NOTE REQUIRED WORK: Write both slice areas in terms of y, compare, then compute the cylinder-minus-cone volume.

Any valid explanation. Essential: both areas as functions of y; Cavalieri's principle; 144π; doubled 288π = 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius r fits exactly inside a cylinder of radius r and height 2r. Use the Cavalieri result for the hemisphere to explain why the sphere has ⅔ of the cylinder's volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

3867

QUESTION ID

S96-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each hemisphere has volume (cylinder of height r) − (cone) = $\pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}$ of πr^3; doubling, the sphere is ⅔ of the cylinder $2\pi r^3$: 4/3 πr^3.

GRADING NOTE REQUIRED WORK: Split the cylinder at the sphere's center into two cylinders of height r and compare each with a hemisphere.

Any valid explanation. Essential: two halves; each hemisphere is ⅔ of its half-cylinder by the Cavalieri comparison; ⅔ of the whole (Archimedes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain, using Cavalieri's principle, why an oblique cylinder of radius r and vertical height h has the same volume as a right cylinder of radius r and height h.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

3868

QUESTION ID

S96-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Every horizontal slice of both is a disk of area πr^2, and the heights are equal, so the volumes are equal: $\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Compare cross-sections at every height.

Any valid explanation. Essential: equal slice areas at every level; equal heights; Cavalieri's principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

A student turns the removed cone upright (base on the bottom of the cylinder, vertex at the top) and compares the remaining solid with the hemisphere. Explain why Cavalieri's principle cannot be used for this comparison, by comparing the two cross-section areas at height y = 0.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

3869

QUESTION ID

S96-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

At y = 0 the hemisphere slice has area πr², but the cylinder slice is completely removed by the cone base (area πr² − πr² = 0); the areas differ, so the principle does not apply.

GRADING NOTE

REQUIRED WORK: Compute both slice areas at y = 0.

Any valid explanation. Essential: both areas at y = 0; unequal areas at some level; conclusion that the hypothesis of Cavalieri's principle fails.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Solids with equal heights and equal cross-section areas at every level have equal volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

3871

QUESTION ID

S96-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

3872

QUESTION ID

S96-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

A prism and a cylinder both have base area 30 and height 4. Volumes?

A. Cannot tell

B. 30 and 4

C. 120 and 40

D. Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

3873

QUESTION ID

S96-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both 120

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

3874

QUESTION ID

S96-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The heights and the cross-section areas at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

True or false?

Cavalieri's principle needs the cross-sections to be the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

3875

QUESTION ID
S96-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

3877

QUESTION ID

S96-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Find the radius of the horizontal slice at height y.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

3878

QUESTION ID

S96-B2-Q2

TYPE

Short answer

CORRECT ANSWER

√(r² − y²)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

A. 4

B. 8

C. 16

D. √136

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

3879

QUESTION ID

S96-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 100 − 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

True or false?

The slice radius, the height, and the sphere radius form a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

3880

QUESTION ID

S96-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angle at the slice center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

True or false?

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Its horizontal slices get bigger as y increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

3881

QUESTION ID

S96-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

3883

QUESTION ID

S96-B3-Q1

TYPE

Short answer

CORRECT ANSWER

216π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

3884

QUESTION ID

S96-B3-Q2

TYPE

Short answer

CORRECT ANSWER

72π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder minus cone (same r and h = r) =

A. $4/3\ \pi r^3$

B. πr^3

C. $\frac{2}{3}\pi r^3$

D. $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

3885

QUESTION ID

S96-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{2}{3}\pi r^3$

GRADING NOTE $\pi r^3 - \frac{1}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A cone is ⅓ of the cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

3886

QUESTION ID
S96-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

3887

QUESTION ID

S96-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{2}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

3889

QUESTION ID

S96-B4-Q1

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

True or false?

Ring area = $\pi(R^2 - r^2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

3890

QUESTION ID

S96-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer 4 and inner 1:

A. 3π

B. 15π

C. 9π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

3891

QUESTION ID

S96-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15π

GRADING NOTE 16 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

3892

QUESTION ID

S96-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\pi(r^2 - y^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

True or false?

Ring area = $\pi(R - r)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

3893

QUESTION ID

S96-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\pi(R^2 - r^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

3895

QUESTION ID

S96-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 5,000 m³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

3896

QUESTION ID

S96-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Equal: 120 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

3897

QUESTION ID
S96-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equal slices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

A. 40 and 13.3

B. 5 each

C. Both 40

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

3898

QUESTION ID

S96-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Both 40

GRADING NOTE 5 × 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

3899

QUESTION ID

S96-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the areas must be equal at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

3901

QUESTION ID

S96-F2-Q1

TYPE

Short answer

CORRECT ANSWER

298.7 m³

GRADING NOTE Rounding to 0.1 required. $\frac{2}{3}\pi(10^3 - 9.5^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

3902

QUESTION ID

S96-F2-Q2

TYPE

Short answer

CORRECT ANSWER

717 t

GRADING NOTE Rounding to the nearest ton required. Volume $\frac{2}{3}\pi(10^3 - 9.5^3) \approx 298.71$ m³; × 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

True or false?

A shell's volume is the outer volume minus the inner volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3903

QUESTION ID

S96-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3903

A hemisphere of radius 3 has volume

A. 9π

B. 27π

C. 36π

D. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3904

QUESTION ID

S96-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 18π

GRADING NOTE ⅔π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3904

ANSWER KEY Reduced copy of the question page, with its answer

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3905

QUESTION ID
S96-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Volumes subtract, not radii: the shell is the difference of two hemispheres.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3905